

Advanced (Habanero)

Q1. Simplify $2^3 \cdot 2^4$

- (A) 2^{3+4}
- (B) $2^{3 \cdot 4}$
- (C) 2^{3-4}
- (D) $2^{3/4}$
- (E) 2^{4-3}

Q2. Which rule applies to $(a^m)^n$?

- (A) Product rule
- (B) Quotient rule
- (C) Power rule
- (D) Sum rule
- (E) Difference rule

Q3. Simplify

- $a^8 a^3$
- (A) a^{8+3}
 - (B) a^{8-3}
 - (C) $a^{8 \cdot 3}$
 - (D) $a^{8/3}$
 - (E) a^{3-8}

Q4. Simplify $3^2 \cdot 3^5$

- (A) 3^{2+5}
- (B) $3^{2 \cdot 5}$
- (C) 3^{2-5}
- (D) $3^{2/5}$
- (E) 3^{5-2}

Q5. Simplify $(2^2)^3$

- (A) 2^{2+3}
- (B) $2^{2 \cdot 3}$
- (C) 2^{2-3}
- (D) $2^{2/3}$
- (E) 2^{3-2}

Q6. Simplify

- $2^8 2^2$
- (A) 2^{8+2}
 - (B) 2^{8-2}
 - (C) $2^{8 \cdot 2}$
 - (D) $2^{8/2}$
 - (E) 2^{2-8}

Q7. Simplify $(a^2)^4 \cdot a^3$

- (A) $a^{2 \cdot 4+3}$
- (B) $a^{2 \cdot 4-3}$
- (C) $a^{2+4 \cdot 3}$
- (D) $a^{2+3 \cdot 4}$
- (E) $a^{4 \cdot 2+3}$

Q8. Simplify

- $3^6 3^2$
- (A) 3^{6+2}
 - (B) 3^{6-2}
 - (C) $3^{6 \cdot 2}$
 - (D) $3^{6/2}$
 - (E) 3^{2-6}

Open Ended Questions (FRQ)

Q9. Simplify $(2a^3)^2 \cdot (3a^2)^4$ and show all steps.

Q10. If $a^m \cdot a^n = a^{m+n}$, prove that $(a^m)^n = a^{m \cdot n}$ using the given equation and show all steps.

Chef's Tips

- Read each question carefully.
- Use the appropriate exponent rule for each question.
- Show all steps for free response questions.